

1. PROJECT OVERVIEW

The Sales Data Analysis project uses Python and the Pandas library to analyze a sales dataset containing 100 sales records and 5 columns.

The purpose of this project is to practice working with real-world structured data and understand the basic data analysis workflow.

Objectives

- Load sales data from a CSV file.
- Explore the structure and contents of the dataset.
- Identify missing values and duplicate records.
- Clean the dataset.
- Calculate important sales metrics.
- Identify the best-selling product.
- Generate a formatted sales analysis report

2. DATASET DESCRIPTION

The dataset contains the following columns:

Column	Description
Date	Date on which the sale occurred
Product	Name of the product
Category	Product category
Quantity	Number of units sold
Total Sales	Total revenue generated from the sale

3. TOOLS AND TECHNOLOGIES

- Python 3
- Pandas
- CSV
- GitHub

The project does not require any external libraries other than Pandas

4. PROJECT STRUCTURES

Week-03/

- |— sales_analysis.py
- |— sales_data.csv
- |— analysis_report.md
- |— requirements.txt

5. SETUP INSTUCTIONS

Step 1: Install Python

Install Python 3 on your computer.

Step 2: Open the Project

Open the Week-03 folder using VS Code or a terminal.

Step 3: Install Pandas

Run:

```
pip install -r requirements.txt
```

Alternatively:

```
python -m pip install pandas
```

Step 4: Run the Program

Run:

```
python sales_analysis.py
```

On Windows, the following command can also be used:

```
py sales_analysis.py
```

```
best_product = product_sales.idxmax()
```

This provides the best-selling product based on total revenue.

6. TECHNICAL DETAILS

Data Structure

Pandas DataFrame is the primary data structure used in this project.

A DataFrame stores data in rows and columns and provides functions for filtering, grouping, cleaning, and analyzing data.

Algorithms Used

The project uses:

1. CSV data loading.
2. Missing-value detection.
3. Median-based missing-value replacement.
4. Duplicate detection and removal.
5. Sum calculations.
6. Mean calculations.
7. Maximum and minimum calculations.
8. Grouping by product.
9. Sorting product performance

7. TESTING

The program was tested using the following scenarios:

Test	Expected Result
CSV loads successfully	Dataset displayed
Check shape	Number of rows and columns displayed
Missing values	Missing values identified
Duplicate records	Duplicate records removed
Total revenue	Correct total calculated
Average sales	Correct average calculated
Highest sale	Maximum sale identified
Lowest sale	Minimum sale identified
Product analysis	Products ranked by revenue

8. CONCLUSION

This project demonstrates the basic workflow of data analysis using Python and Pandas.

The workflow followed was:

Load Data



Explore Data



Identify Problems



Clean Data



Analyze Data



Generate Report

Through this project, fundamental data-analysis concepts such as DataFrames, CSV files, missing-value handling, duplicate removal, aggregation, grouping, and descriptive statistics were practiced.